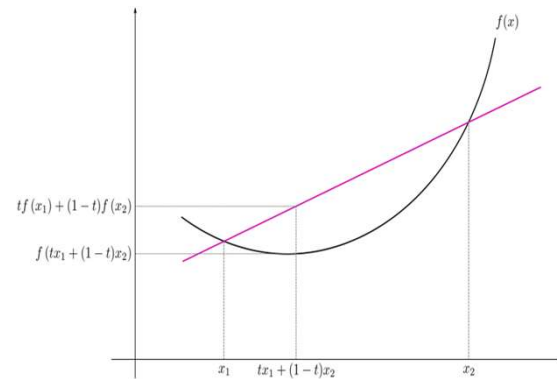


Agenda

- Last time
 - Weak Law of “large numbers”
 - Popular distribution functions



2

Topic Overview

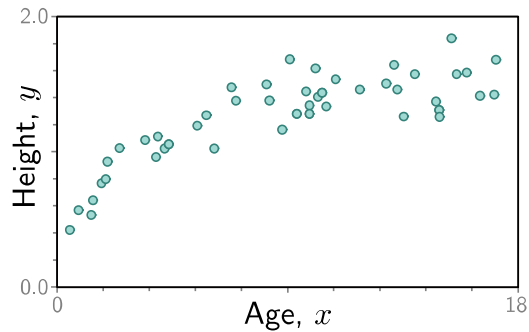
- We wish to study several useful families of random variables
 - Discrete random variables: Bernoulli, Categorical, Binomial, Poisson, hypergeometric
 - Continuous random variables: Uniform, Gaussian, exponential, chi-square
- This may become tedious, so why?

2

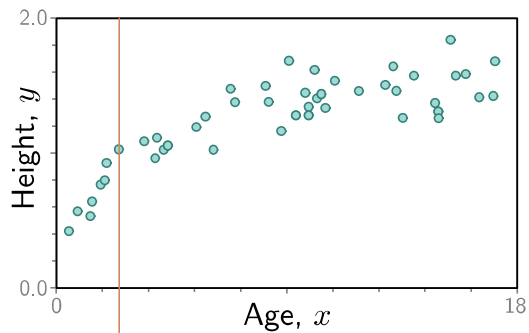
3

Finding the trend

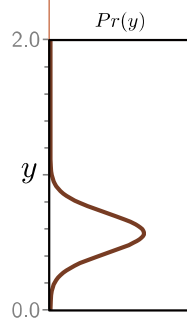
- You may have studied how to find the line fitting given points in your earlier classes



4



However, the output is often not a specific line (a y -coordinate for a given x) but a distribution.

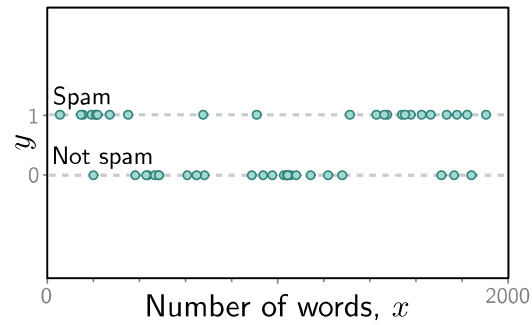


What is the nature of the distribution?

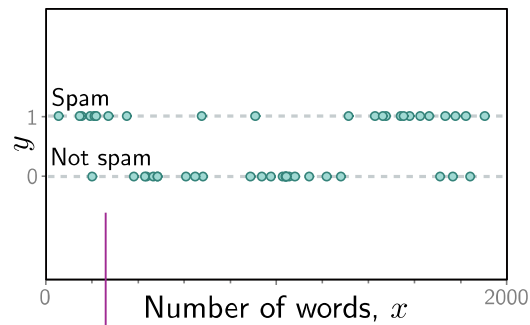
5

Text classification

- Is the email spam or not?
- We may want a probabilistic interpretation of the output



6



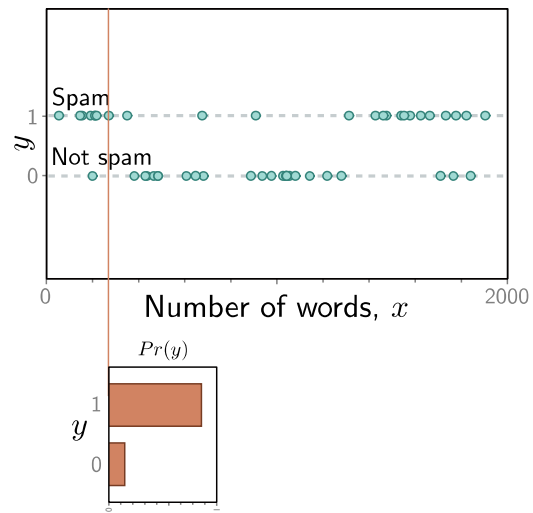
Text Classification

What is the nature of the output distribution?



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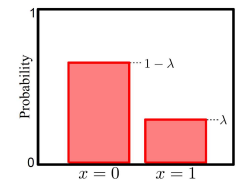


8

Bernoulli Distribution

- Example: Biased coin producing a heads with probability λ
- Outcome is either a success or a failure
- Single parameter
- Properties
 - Expectation: p
 - Variance: $p(1-p)$

Must know



$$Pr(x=0) = 1 - \lambda$$

$$Pr(x=1) = \lambda.$$

$$Pr(x) = \text{Bern}_x[\lambda]$$

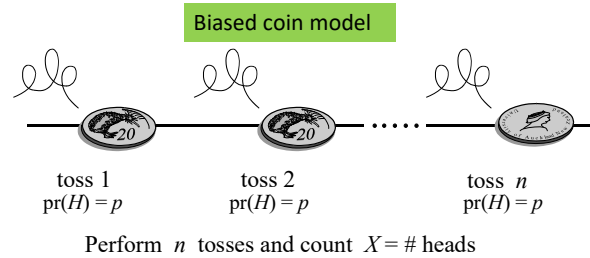
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9

Binomial distribution

Physical model for situations of a series of trials

- Each trial has only two outcomes: success and failure;
- $p = \text{pr}(\text{success})$ is the same for every trial; and
- trials are independent



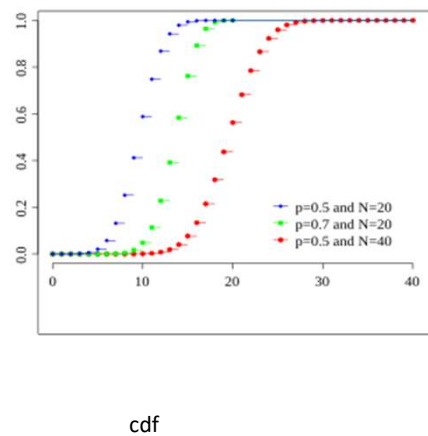
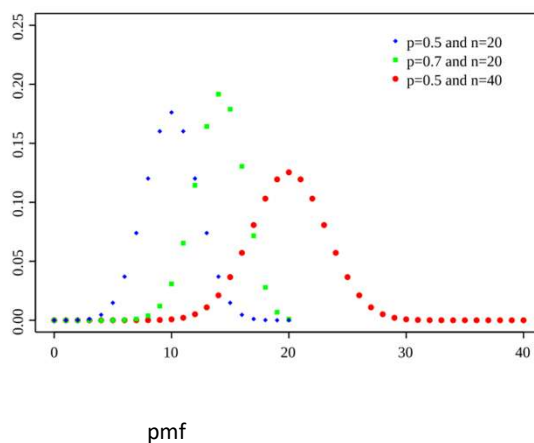
$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$$

$$\sum_i P(X = i) = \sum_i \binom{n}{i} p^i (1 - p)^{n-i} = (p + 1 - p)^n$$

Binomial
Theorem

12

Binomial distribution



13

Urn model

- If we sample Y individuals at random from a finite population and
- Count X , the number of individuals with a characteristic of interest
 - Example: IITB is good about training students for the next generation. Agree or Disagree?
- What can we say about the characteristic?
- Is the distribution of X binomial?
 - The distribution of X is approximately Binomial($Y, p=M/N$).
 - Operating Rule: $Y/N < 0.1$

Useful
approximation

N balls in an urn, of which there are

M black balls

$N - M$ white balls

